

Product Requirement Document (PRD)

App Name

PALASH-Setu

Tagline

An offline-first, voice-powered AI co-pilot enabling Hindi-medium primary teachers to deliver mother-tongue education in Ho, Mundari, and Santhali within 3 seconds.

Problem

In over 5,000 tribal-area primary schools across Jharkhand, foundational literacy and numeracy (FLN) is severely bottlenecked under the PALASH MTB-MLE programme. The vast majority of assigned primary teachers are Hindi-medium trained and cannot speak or read local tribal languages (Ho, Mundari, Santhali). Consequently, Grade 1–5 tribal children are taught in a language they do not comprehend at home, leading to low comprehension, high dropout rates, and learning loss. Existing digital translation tools fail because they require active internet, lack tribal script support, are not aligned with FLN pedagogy, and crash on low-cost government tablets (≤ 2 GB RAM, Android 9+).

Target User

The primary user is a non-native, Hindi-medium government primary school teacher assigned to rural or tribal-belt schools in Jharkhand. They need a zero-friction, voice-first teaching assistant on low-end tablets to communicate with tribal students, create localized lesson plans, and generate bilingual learning materials without prior language training.

Core Features (Must Have)

- **Sub-3s Voice-to-Voice Classroom Translator:** Low-latency Hindi speech input converted instantly to synthesized tribal speech (Santhali, Ho, Mundari) and on-screen dual-script text.
- **Dual-Script Real-Time Display:** Simultaneous rendering of target native tribal scripts (Ol Chiki, Warang Chiti) for students alongside Devanagari transliteration for Hindi teachers.

- **100% On-Device Offline Engine:** Quantized lightweight pipeline (Vosk Hindi ASR + INT4 IndicTrans2 NMT + Piper TTS running via C++ ONNX Runtime) operating strictly under 600MB RAM without internet.
- **FLN Hybrid Lexicon Lookup:** Deterministic local dictionary fallback (<5ms) for NIPUN Bharat core vocabulary (numbers, shapes, basic commands) to prevent neural model hallucinations.
- **Bilingual Worksheet & Flashcard Generator:** On-device vector PDF renderer producing NIPUN FLN-aligned bilingual exercise sheets and visual flashcards on demand.
- **FLN Lesson & Pedagogy Script Generator:** Pre-loaded template engine converting Hindi FLN teaching goals into step-by-step vernacular activity scripts and local story prompts.

Nice to Have

- **Multi-Dialect Region Selector:** Quick toggle for regional sub-dialects of Santhali and Mundari spoken in specific Jharkhand districts.
- **Audio Speed & Pitch Control:** Adjustable speech synthesis speed to slow down pronunciations for early Grade 1 learners.
- **Worksheet Print/Bluetooth Share:** Direct peer-to-peer Bluetooth/Wi-Fi Direct sharing of generated worksheet PDFs between teacher tablets.
- **Offline Usage Analytics:** Local tracking of translated topics and generated worksheets to sync once device reaches network coverage.

Out of Scope

- **Direct Student 1:1 Login Portal:** No independent student user profiles or student-facing gamified dashboards (focus remains strictly on teacher-assisted delivery due to shared classroom tablet constraints).
- **Cloud-Dependent LLM Integration:** No live API calls to external cloud models (GPT-4, Claude, external web APIs) during classroom operation.
- **Higher Education / Secondary School Subjects:** Content is strictly restricted to Primary Grade 1–5 Foundational Literacy and Numeracy (FLN).

User Stories

- **As a** Hindi-speaking primary teacher, **I want to** tap a button and speak a classroom instruction in Hindi, **so that** the app immediately plays the translated audio in Santhali and shows me the Devanagari pronunciation to read out loud.

- **As a non-native teacher in a remote tribal school with no internet, I want** the entire translation and voice pipeline to run fully offline on my low-cost 2GB RAM tablet, **so that** my teaching flow is never interrupted by connectivity drops.
- **As a primary school teacher, I want to** generate a bilingual Hindi-Santhali math worksheet with visual icons, **so that** I can print or display it for my Grade 2 students during activity time.
- **As a teacher introducing new foundational concepts, I want to** look up standard FLN terms (numbers, fruits, body parts), **so that** I am 100% confident the translated tribal terms are accurate and free from AI hallucinations.

Success Metrics

- **Voice-to-Voice Latency:** End-to-end processing time from speech end to audio playback < 1.8 seconds (comfortably under the 3-second PS 42 threshold).
- **Hardware Footprint:** Total application memory usage < 600MB RAM on Android 9+ / API Level 28 devices.
- **Translation Accuracy:** 100% precision on core NIPUN FLN dictionary terms and > 85% BLEU accuracy on contextual classroom dialogue.
- **Worksheet Generation Speed:** On-device PDF render time < 2 seconds for full bilingual flashcard sets.

Technical Requirement Document (TRD)

Document 02

Section	Technical Details & Specification
Frontend	Android Native (Kotlin) with Jetpack Compose / Material Design 3 UI; alternative stripped Flutter build with minimal engine footprint (<150MB total app size).
Backend	On-Device Local C++ Orchestration Layer via JNI (Java Native Interface). Zero server backend needed for core real-time operations. Optional Python/FastAPI microservice used strictly for initial model quantization and data pre-sync scripts.
Database	SQLite via Android Room ORM (Pre-seeded local database for FLN dictionary, mapped NIPUN terms, cached translation pairs, and localized activity templates).
Auth	Local Device PIN / Teacher Device ID Auth via Android EncryptedSharedPreferences (Zero cloud dependencies; works 100% offline).
Hosting	On-Device Offline Execution (Embedded APK / Android System Image). Model weights hosted on GitHub Releases / Hugging Face Hub for one-time initial download/sync over Wi-Fi.
Third-party APIs	• BHASHINI / AI4Bharat IndicTrans2 APIs (Initial fine-tuning/data collection in cloud; Free/Open-Source) • Vosk ASR Engine (Offline speech recognition; Free/Apache 2.0) • Piper TTS Engine (Offline fast neural speech synthesis; Free/MIT)
Key Libraries	• ONNX Runtime Mobile (C++ JNI) — Low-latency on-device neural model inference

	• Tencent NCNN — Highly optimized ARM CPU neural execution framework • Android PdfDocument / Canvas API — Local vector PDF worksheet generator • Android AudioRecord & AudioTrack API — Low-latency PCM 16kHz audio streaming • FreeType / Noto Sans Fonts — Native font rendering for Ol Chiki, Warang Chiti, and Devanagari scripts
Environment Variables	• MODEL_PATH_VOSK_HI • MODEL_PATH_INDICTRANS_ONNX • MODEL_PATH_PIPER_TTS • SQLITE_DB_NAME • MAX_INFERENCE_MEMORY_MB
Constraints	• Strict Memory Footprint: Total runtime RAM usage must stay under 600 MB to avoid OS termination on 2GB RAM Android 9+ tablets. • Strict Latency: End-to-end voice-to-voice translation loop (ASR -> NMT -> TTS) must execute in < 3.0 seconds (Target: 1.5s–1.8s). • Offline Operation: Zero reliance on cellular/Wi-Fi networks post-sync. • Target Hardware: Low-cost ARM-based Android tablets (Snapdragon 400 series / MediaTek Helio, Android 9 / API Level 28+).

App Flow Document

Document 03

Section	Specifications & Application Flow
Pages List	• /onboarding (Language & School Setup) • /live-translator (Core Voice-to-Voice Classroom Dashboard) • /fln-scripts (Lesson Plan & Vernacular Activity Prompts) • /worksheet-generator (Bilingual Worksheet & Flashcard Generator) • /dictionary (FLN Dictionary Lookup & Term Verification) • /settings (Offline Storage, Dialect Selection & Model Diagnostics)
Navigation Type	Bottom Navigation Bar (3 Primary Tabs: Live Translator, FLN Scripts, Worksheets) + Top Action Header (Language Pair Toggle, Offline Status Indicator, Settings Icon).
First Screen	/onboarding (Language Setup Screen) — A clean, low-literacy-friendly welcome screen allowing the teacher to select their preferred target tribal language (Santhali, Ho, or Mundari) and district/block name.
Auth Flow	Offline Setup: Launch App $\rightarrow$ Device Authorization / School Code Entry $\rightarrow$ Select Target Tribal Language $\rightarrow$ Redirect to /live-translator (Zero cloud authentication or network verification needed).

Core User Journey 1	Real-Time Voice Translation in Classroom: 1. Teacher opens app $\rightarrow$ Automatically lands on /live-translator. 2. Teacher taps the prominent Push-to-Talk Mic button and speaks a classroom command in Hindi (e.g., "बच्चों, अपनी गणित की किताब का पन्ना नंबर 5 खोलो"). 3. Audio is captured and processed on-device via Vosk ASR $\rightarrow$ IndicTrans2 NMT $\rightarrow$ Piper TTS. 4. In < 1.8 seconds, the native Santhali/Ho audio plays automatically through device speakers. 5. Screen displays the translated text in Native Script (Ol Chiki) for students and Devanagari Transliteration for the teacher to read out loud.
Core User Journey 2	Generating a Bilingual FLN Worksheet: 1. Teacher navigates to /worksheet-generator via the bottom navigation bar. 2. Selects NIPUN FLN competency (e.g., <i>Grade 1 Math — Counting 1 to 10</i>) and output format (<i>Worksheet</i> or <i>Flashcards</i>). 3. Taps "Generate PDF". 4. On-device Android Vector Engine generates a bilingual Hindi-Santhali PDF with visual illustrations in < 2 seconds. 5. Teacher previews the PDF directly on screen, saves it locally, or shares it via Bluetooth/Wi-Fi Direct to a local printer.
Empty States	• No Voice Input Yet (/live-translator): Displays a friendly wave-animation prompt with sample spoken Hindi phrases ("Tap the mic and say a command...").

	• No Generated Worksheets (/worksheet-generator): Displays a graphic illustration with a "Create Your First Worksheet" call-to-action button. • No Search Results (/dictionary): Displays "Term not found in standard FLN dictionary — Neural NMT will handle this word dynamically."
Error States	• Audio Microphone Permission Denied: Redirects to a visual permission helper screen with a single "Grant Microphone Access" button. • Low Storage Warning (<100MB Free): Top banner alert: "Storage low — Clear cached PDFs to ensure smooth voice translation." • System Resource Pressure (RAM Limit Exceeded): Automatically frees background audio buffers and falls back to text-only mode with a non-intrusive alert.
Redirects	• First Launch $\rightarrow$ /onboarding • Post-Onboarding / Subsequent Launches $\rightarrow$ /live-translator • Tap "Generate Script" on Dictionary Term $\rightarrow$ /fln-scripts • Tap "Settings" $\rightarrow$ /settings

UI/UX Design Brief Document

Document 04

Section	Specifications & Design System
Aesthetic	Clean, High-Contrast, Educational & Accessible. Designed specifically for sunlight-readable classroom tablets operated by non-native teachers. Large touch targets, low-literacy-friendly visual iconography, and intuitive dual-script cards.
Primary Color	#0066CC (Palash Deep Blue) — Authoritative, professional, and accessible for primary education interfaces.
Background Color	#F8F9FA (Off-White / Light Gray) — Reduces eye strain under bright outdoor/classroom lighting while conserving screen battery on LCD tablets.
Text Color	#1A1D20 (Charcoal / Near-Black) — Provides crisp contrast ratios (>7:1) for maximum legibility of Devanagari, Ol Chiki, and Warang Chiti typography.
Accent / CTA Color	#D9381E (Palash Flame Red) for primary voice recording mic actions; #00875A (Forest Green) for auto-play indicators and successful generation triggers.
Font	• Noto Sans Devanagari (for Hindi text and teacher transliterations) • Noto Sans Ol Chiki (for Santhali text rendering) • Noto Sans Warang Chiti (for Ho text rendering) • Inter / System UI (for navigation labels, buttons, and system metrics)

Border Radius	12px rounded corners throughout cards and visual container blocks for a modern, approachable feel. 50% fully rounded circular borders for the central push-to-talk mic button.
Shadows	0px 2px 8px rgba(0, 0, 0, 0.08) — Subtle card elevation to distinguish translated output cards and worksheet preview tiles without rendering overhead on low-end GPUs.
Dark/Light Mode	Light Mode Primary. High-contrast light mode is default for daytime classroom visibility. Dark mode (#121212 background, #E0E0E0 text) is optional for battery-saving mode on low battery.
Reference Apps	• Google Read Along (Bolo) — For child-friendly, high-contrast, speech-first interaction layout. • Duolingo — For clean, card-based bilingual text displays and clear micro-interactions. • DIKSHA App — For familiar Indian government educational layout patterns.
Mobile	Optimized for 7-inch to 10-inch Android Tablets in landscape/portrait orientation. Features a fixed bottom navigation bar, persistent top language-selector header, and prominent central microphone control.

Backend & Database Schema Document

Document 05

Section	Specifications & Database Schema
Table: users	id (text, primary key), school_code (text), teacher_name (text), district (text), block (text), preferred_target_language (text), created_at (timestamp), role (text)
Table: fln_dictionary	id (integer, primary key), hindi_term (text, indexed), santhali_ol_chiki (text), santhali_devanagari (text), ho_warang_chiti (text), ho_devanagari (text), mundari_devanagari (text), domain_category (text — e.g., 'math_numbers', 'animals', 'body_parts'), grade_level (integer)
Table: translation_history	id (uuid, primary key), user_id (FK $\rightarrow$ users.id), source_hindi_text (text), target_language (text), translated_native_text (text), translated_devanagari_text (text), audio_duration_ms (integer), processing_latency_ms (integer), is_dictionary_exact_match (boolean), created_at (timestamp)
Table: generated_worksheets	id (uuid, primary key), user_id (FK $\rightarrow$ users.id), title (text), target_language (text), nipun_competency_code (text), pdf_local_file_path (text), created_at (timestamp)
Relationships	translation_history.user_id $\rightarrow$ users.id (many-to-one)

	• <code>generated_worksheets.user_id</code> $\rightarrow$ <code>users.id</code> (many-to-one)
Auth Provider	Local Device Storage Auth (Offline First) — Android EncryptedSharedPreferences stores teacher profile & local device ID. Zero external server auth needed for offline operation. Optional sync via JWT tokens when internet connectivity becomes available.
Row Level Security	SQLite Local Access Control — Local SQLite database operates under standard Android Sandboxed App Storage (<code>/data/data/com.palash.setu/databases/</code>), ensuring only the PALASH-Setu app can read/write database rows.
User Roles	• teacher (Default: full access to translation, worksheet generation, and local dictionary search) • school_admin (Access to offline usage diagnostics and bulk sync settings)
File Storage	Android Internal Scoped Storage — Generated PDFs stored in <code>/storage/emulated/0/Android/data/com.palash.setu/files/worksheets/</code> ; Local speech audio buffers cached in <code>/cache/audio_temp/</code> and purged after playback.
Sensitive Fields	• <code>school_code</code> & <code>teacher_name</code> — Encrypted locally using AES-256 via Android Master KeyStore. No child or student personal identifiable information (PII) is captured or stored in the database.

Implementation Plan Document

Document 06

Section	Development Phases & Technical Milestones
Phase 1: Setup	• Init native Android project (Kotlin) / stripped Flutter codebase; configure C++ JNI bridge directory (/app/src/main/cpp). • Set up repo structure (android/, models/, scripts/). • Configure build environment variables (MODEL_PATH_VOSK_HI, MAX_INFERENCE_MEMORY_MB) and install core dependencies (onnxruntime-android, vosk-android, room-runtime).
Phase 2: Database	• Create SQLite Room database schemas for users, fln_dictionary, translation_history, and generated_worksheets. • Seed local DB with 1,500+ NIPUN FLN domain terms (Santhali/Ol Chiki, Ho/Warang Chiti, Mundari, Devanagari transliterations). • Implement Room DAO interfaces and write local migration scripts.
Phase 3: Auth	• Implement offline local setup flow (/onboarding) saving teacher profile, district, and target language preference to EncryptedSharedPreferences.

	• Set up session state manager to auto-bypass setup on subsequent launches and route directly to /live-translator.
Phase 4: Core Feature 1	Sub-3s On-Device Real-Time Voice-to-Voice Translator: 1. Integrate AudioRecord API to stream PCM 16kHz audio to C++ layer. 2. Bind Vosk-Hindi ASR (vosk-model-small-hi-0.22) for sub-400ms speech-to-text. 3. Implement hybrid translation pipeline: Fast SQLite dictionary intercept (<5ms) $\rightarrow$ INT4 Quantized IndicTrans2 ONNX fallback (~600ms). 4. Bind Piper TTS engine for Santhali/Ho audio synthesis (~500ms). 5. Build dual-script UI card rendering Native Script (Ol Chiki/Warang Chiti) + Devanagari Transliteration.
Phase 5: Core Feature 2	Bilingual NIPUN FLN Worksheet & Flashcard Generator: 1. Implement template selection engine for Grade 1–5 FLN competencies (Math, Phonics, Vocabulary). 2. Integrate Android PdfDocument / Vector Canvas API. 3. Embed Noto Sans Ol Chiki & Devanagari fonts for local rendering without external asset fetching.

	4. Build PDF preview tile UI, local storage saver, and Bluetooth/Wi-Fi Direct sharing handler.
Phase 6: Core Feature 3	FLN Lesson Script & Vernacular Activity Generator: 1. Create /fln-scripts dashboard with dropdown selectors for NIPUN FLN topics. 2. Map Hindi pedagogical prompts to local cultural activity templates and story prompts. 3. Build offline dictionary lookup tool (/dictionary) with fuzzy search for instant FLN term validation.
Phase 7: UI Polish	• Apply Palash UI design system (Deep Blue #0066CC, Off-White #F8F9FA, Palash Red #D9381E). • Build micro-interaction animations (pulsing mic visualizer during speech recording, progress indicator during PDF generation). • Implement high-contrast light mode layout, empty states, and permission helper screens.
Phase 8: Testing	• Perform end-to-end latency benchmarks on budget 2GB RAM / Android 9+ target hardware (verify voice-to-voice loop completes in $< 1.8s$). • Conduct memory stress testing using Android Studio Profiler (verify total heap allocation stays strictly $< 600MB$).

	• Test 100% offline flight mode execution and edge case recovery (mic permission denial, low storage, background audio clearing).
Phase 9: Deploy	• Compile production-ready standalone APK with bundled quantized models (.onnx assets). • Upload APK and source code to GitHub repository. • Record 3-minute video demonstration on a 2GB RAM budget Android tablet showing offline voice-to-voice translation and worksheet generation.
Done Criteria	Application is fully functional end-to-end on a budget 2GB RAM Android 9+ tablet in Airplane Mode (No Internet): 1. Hindi teacher speaks into mic $\rightarrow$ Santhali audio plays back in < 1.8 seconds. 2. Dual-script (Ol Chiki + Devanagari) displays accurately on screen. 3. Bilingual NIPUN FLN worksheet PDF generates locally in < 2 seconds. 4. Peak memory usage remains strictly < 600 MB with zero system crashes (OutOfMemoryError).

